

Amjad S Althabteh

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Education

B.S in Computer Science	GPA: 3.55/4.00	Wayne State University	Detroit	2026
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Experience

Co founder	Immersalab Graphics	Detroit, MI	Sep 2025 - Dec 2025
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- Architected a real-time rendering engine in **C++20 and OpenGL**, focusing on **GPU-driven rendering**.
- Robust internal UI and asset-authoring tools built on JSON-driven pipelines, greatly expanding the flexibility of content creation.
- **Data-driven rendering optimizations** to allow runtime analysis and AI guided heuristics to balance visual quality, GPU performance.
- Modular rendering subsystems with clear ownership boundaries to support iteration and feature growth
- GPU-side profiling and frame-timing instrumentation to guide optimization and debugging.

Unreal C++ Dev	Indie Game AAA	Ann Arbor, MI	Apr 2025 - Aug 2025
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- Wrote gameplay systems using(**C++ 20 and Visual Scripting**), connecting animation states, environmental triggers, and narrative logic to create seamless transitions between cinematic moments.
- Created 3d environments and characters from scratch, rigging for animation, and enhanced each scene.
- Produced a 90+ second real-time cinematic using MetaHumans, Control Rig, and Sequencer designing the flow.
- Dynamic animation-control layer using **C++ and Animation Blueprints**, making smooth blending across all aspects.
- Implemented event-driven state transitions to synchronize gameplay logic, animation, and cinematic systems

Software Development Intern	Rare Munchies	Ann Arbor, MI	Sep 2023 - Apr 2024
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- Integrated UI components using **TypeScript**, improving page transitions and interaction flow to enhance user experience.
- Identified and resolved **cross-browser compatibility issues**, reducing debugging time and unblocking delivery
- Resolved **cross-browser compatibility issues**, reducing debugging overhead and stabilizing releases

Projects

Glint 3D – C++ Automated 3D Rendering Engine

[GlintRender](#)

- Rendering system built on **C++17/OpenGL rendering pipeline with dynamic GPU workload** and lighting optimization.
- Designed a cross-platform architecture with runtime scene analysis to maintain stable performance in complex scenes
- Set up a **modular engine architecture** separating rendering, allowing faster iterations.
- Implemented a **custom memory pool and allocator system** using C++20 features to reduce allocations, + runtime performance.

UE -5.6 Cinematic “No One Answers Clip”

[No one answers clip](#)

- Built a AAA solo cinematic in Unreal Engine 5.6 using MetaHuman animation and cinematic cameras, and control;l rigged anims.
- Used Niagara effects and environmental animation to tell the story visually and set the tone for No One Answers

C++ Runtime Stack Trace & Diagnostics Engine (Traceforge)

- Developed a **C++ 23 debugging system** that performs automated stack-trace parsing.
- Implemented automated regression test generation and intelligent fix suggestions to reduce debugging time.
- Added **thread-safe logging** to capture errors and runtime events in multithreaded applications.
- Basic **memory tracking utilities** to help identify leaks and invalid allocations during debugging.

Echos of Sacrifice Godot

[EchosOfSacrifice Slice](#)

- Developed a modular **JSON-based save/load system** in Godot using GDScript, enabling persistent player position, level state, and progress without requiring an external database..
- Integrated event-driven save/load triggers with thread-safe state handling through a centralized GameManager singleton.
- Architected modular scene and script organization in Godot (GDScript), reducing coupling between gameplay logic and UI components.

Incident Management Ops

- Built a real-time incident management platform with **FastAPI + React**, supporting collaboration rooms, async workflows, and Prometheus metrics for production observability
- Developed an end-to-end monitoring pipeline with automated alert ingestion, severity classification.

Skill

- **Programming Languages:** C++ 11 - 26, C#, C, Python 3.7 - 3.12, TypeScript, Java
- **Graphics & Engines:** OpenGL • **Unreal Engine 4 - 5.7** • Unity • Godot • NX Design
- **Course Work:** Computer Graphics, Game Programming, **Software Engineering**, **Operating Systems**, Linear Algebra
- **Specialties:** Gameplay Systems, VR/AR Development, Simulation, Tools & Plugins, Technical Art